

AGGREGATED PLANNING MODEL DOCUMENTATION

The following document develop and explain aggregated planning model with the goal of building a 4-month Supply and demand operational plan for Gel packs that are used for maintaining perishable goods in transit in good condition.

- AGGREGATED UNIT: Pallets
- RESOURCES: Trucks

SETS OF INDEXES

- $t \in T = \{1, 2, \dots, T\}$: time periods (weeks, e.g. 16 weeks)
- $k \in K = \{\text{Type A (64 oz), Type B (32 oz)}\}$
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DEFINITION OF MODEL VARIABLES

- Let P_t^k be the number of pallets ordered of product k in the period t .
- Let I_t^k be the frozen and ready inventory of product k at the end of period t .
- Let B_t^k be the backorder of frozen and ready inventory of product k at the end of period t .
- Let F_t^k be the freezer inventory (WIP) of product k at the end of period t .
- Let D_t^k be the demand of product k at the end of period t .
- Let W_t be the number of trucks to be used for delivery in period t .
- Let L_t be the number of trucks deactivated at period t .
- Let H_t be the number of trucks activated at period t .

DEFINITION OF MODEL COSTS VARIABLES

- Let $C_t^{P,k}$ be the unit purchasing cost per pallet of product k ordered in period t .
- Let $C_t^{I,k}$ be the unit inventory holding cost per pallet of frozen and ready inventory of product k held during period t .
- Let $C_t^{F,k}$ be the unit inventory holding cost per pallet of freezer work-in-process (WIP) inventory of product k held during period t .
- Let $C_t^{B,k}$ be the unit backorder penalty cost per pallet of unmet demand of frozen and ready inventory of product k at the end of period t .
- Let C_t^W be the unit transportation cost of using one truck in period t .
- Let C_t^H be the unit cost of activating one truck in period t .
- Let C_t^L be the unit cost of deactivating one truck in period t .

PARAMETERS

- D_t^k : demand of product k in period t
- $w^{64} = 1,600$ lbs per-pallet
- $w^{32} = 1,800$ lbs per-pallet
- Truck capacity = 32,000 lbs
- Freezing lead time = 4 periods

OBJECTIVE FUNCTION

$$\min Z = \sum_{t,k} [\tilde{C}_t^{P,k} P_t^k + C_t^{I,k} I_t^k + C_t^{F,k} F_t^k + C_t^{B,k} B_t^k] + \sum_t [C_t^W W_t + C_t^H H_t + C_t^L L_t]$$

CONSTRAINTS

Transportation Capacity Constraint (Weight-based capacity)

$$1,600 P_t^{64} + 1,800 P_t^{32} \leq 32,000 W_t \forall t$$

Resource Balance Constraint (Transportation Capacity Dynamics)

$$W_t = W_{t-1} + H_t - L_t \forall t$$

Frozen & Ready Inventory and Backorder Balance

$$I_t^k - B_t^k = I_{t-1}^k - B_{t-1}^k + P_{t-4}^k - D_t^k \forall t, k$$

Freezer Work-in-Process (WIP) Inventory Balance (Freezing lead time modeling)

$$F_t^k = F_{t-1}^k + P_t^k - P_{t-4}^k \forall t, k$$

Safety Stock Requirement (Zero stockout policy)

$$I_t^k \geq 2 \cdot D^k \forall t, k$$

Non-negativity Constraints

$$P_t^k, F_t^k, I_t^k, B_t^k, W_t, H_t, L_t \geq 0$$

No Stockouts Constraints

$$B_t^k = 0 \forall t \in T, \forall k \in K$$